National

Operational Excellence Standard (OES) — Customer Premise Equipment (CPE) installation

HOW22893 | MODIFIED 09/10/2026



Summary

This standard will review the general requirements for the installation of Comcast-

owned CPE.

Quick Links

- At a glance
- CPE location consideration
- Limiting potential interference
- CPE installation
- CPE mounting options
  - When wall mounting CPE
  - When rack mounting CPE
- OES document owner, contributors, and approvers
- What's changed in this article

At a glance

This document covers the requirements for CPE installation to deliver the products and services that a customer has subscribed to. This standard provides guidelines for the installation of customer-owned and Comcast-owned CPE. All CPE connected to Comcasts network must be approved by Comcast Corporate Engineering.

Comcast proprietary and confidential

Information in this document is subject to change without notice. This document may contain sensitive confidential proprietary information and trade secret information concerning Comcast. All recipients have an obligation not to disclose to any non-Comcast person or entity the information contained herein. Partial or full disclosure, even inadvertent disclosure, of this information may cause irreparable damage to

Comcast. No part of this document may be reproduced in whole or in part, or stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, for any purpose, without the express written permission of Comcast. If you are not the intended recipient of this document, you are hereby notified that any dissemination, distribution or copying of this document or its contents is strictly prohibited.

Comcast may have patents, patent applications, trademarks, copyrights, or other intellectual property rights covering subject matter in this document. The furnishing of this document does not grant any license to these patents, trademarks, copyrights, or other intellectual property.

CPE location consideration

- Do not place CPE in an enclosed area where the cooling vents are blocked or impede the flow of air through the ventilation openings.
- Install CPE so that its position does not interfere with its proper ventilation. For example, do not place CPE on a soft surface such as a, sofa, rug, or similar surface that could block the ventilation openings.
- Install CPE away from heat sources such as radiators, heat registers and stoves.
- Installation of CPE near other electronic devices is permitted as long as the location provides adequate circulation for the air surrounding the devices.
- Place CPE on a flat surface not prone to vibration or impact.
- CPE shall be installed inside of a premise and where condensation will not occur.
- Do not install CPE near a water source i.e. sink, bath tub, etc.
- Do not install CPE on a switched power outlet.

Limiting potential interference

- Do not place devices with WiFi technology on top of or right next to other devices with WiFi technology. Devices radiating into our device(s) may impact the performance of our device(s), and likewise the performance of the other devices.
- Keep all WiFi technology devices out of cabinets that have doors and out of shelving that is all metal.
- Only place WiFi technology devices in media panels designed for WiFi. These are made of plastic never metal.
- Avoid placing WiFi Technology devices next to mirrors, glass and/or fish tanks.
- Do not place devices with WiFi technology next to large appliances or microwaves.

CPE installation

- Use Comcast-approved cords and cables to complete all connections.
- Verify that the correct power cord/adapter is used for the equipment you are installing.
  - This information can be located on the manufacturer's label on the device.
- Install all wiring in a neat and tidy fashion.
- Ensure that cords are not crossing walkways or pose a tripping hazard.
- Power cords must never be routed/fished through a wall or in a wall cavity.
- Cords are not to be placed under rugs or carpeting.
- Protect the cords from being pinched particularly plugs/wall plates and at the connection points of the device.
- Ensure all RF connectors are tightened to specifications.
- Some CPE may be designed to be mounted on a wall, or in a rack. See the below table regarding equipment type and mounting options.

CPE mounting options

Equipment model	Wall	Mini-rack	Head-end rack	Comments
Xi5	X			Requires cradle
Xi6	X			Requires cradle
XB7/XB8	X			Requires a bracket or shelf for a media panel installation
CBR/CBR2	X		X	Head-end bracket provided separately
AER1650	X			Connection Pro 1 Gateway
E110-C4D	X			Connection Pro 2 Gateway
100	X			WiFi Pro
130	X			WiFi Pro
200	X			WiFi Pro
C-75	X			WiFi Pro

Equipment model	Wall	Mini-rack	Head-end rack	Comments
Juniper SRX320	X	X	X	
Juniper SRX345			X	Rack mountable only/bottom of the rack
Juniper SRX1500			X	Rack mountable only/bottom of the rack
Cisco C1111-8P	X		X	
Cisco ISR4461/K9			X	Requires three rack units
Cisco ISR4331/K9	X		X	
Advantech FWA-1012VC-8C			X	Used for SD router and SDWAN
Advantech FWA-3260			X	

When wall mounting CPE

- Do not mount the device near HVAC outlets, radiators, or any other source of heat.
- Do not mount the device near any object(s) that could restrict air flow to the device.
- All wiring and cables need to be neat and secured.
- Wall anchors must be used for all hollow wall installations.

When rack mounting CPE

- Be sure to follow guidance on rack positioning and required space.
- CPE must be secured properly in the rack.
- All wiring and cables need to be routed properly, secured and labeled.

OES document owner, contributors, and approvers

Document owner:

- James Kolcun, Director of Technical Operations, HQ

Contributors:

- Scott Melanson, Technical Operations, Northeast Division
- Mike Darnell, Technical Operations, Central Division
- James Mascarenas, Technical Operations, West Division

Approvals:

- Meena Soleiman, SVP Tech Ops & Eng: Northeastern Division
- Rachel Solomon, SVP Tech Ops & Eng: Western Division
- Tony Speller, SVP Tech Ops & Eng: Central Division
- Stephen Zanetich, VP Technical Operations: HQ
- Patrick McDonald, VP2 Technical Operations: HQ
- Michael Hewitt, SVP Technical Operations: HQ

What's changed in this article

Date	Summary of change
September 10, 2026	Added 100, 130, and 200 models to CPE mounting options table.
September 4, 2025	Added to Celestial.
August 9, 2024	Conducted annual review and updated: • Removed temperature reference. • Added hollow wall anchor guidance. • Removed wall mount DTA.
September 30, 2022	Conducted annual review and updated.
June 1, 2021	Released document.

↑ Back To Top